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MTH632 Complex Analysis and Differential Geometry

Question No : 6 of 33

If U and V are harmonic functions then $f(Z) = U + i\,V$ is _____.

Answer (Please select your correct option)

- analytic function.
- need not be analytic function.
- analytic function only at $Z=0$.
- none of the others.

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 22 of 37

Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v

Answer (Please select your correct option)

- C $(\phi+\psi)(v)$
- C $(\phi+\psi)(v)=(\psi)(v)$
- C $(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
- C $(\phi+\psi)(v)=(\phi)(v)$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 13 of 33

The circle $z = e^{-i\theta}$, $0 \le \theta \le 2\pi$ is oriented in _____ direction.

Answer (Please select your correct option)

- upward
- downward
- counter-clockwise
- clockwise

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 22 of 33

Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v

Answer (Please select your correct option)

- C $(\phi+\psi)(v)$
- C $(\phi+\psi)(v)=(\psi)(v)$
- C $(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
- C $(\phi+\psi)(v)=(\phi)(v)$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $arcsin\ (Z)$ = _____.

Answer (Please select your correct option)

- $\log(iz + (1-z^2)^{1/2})$
- $-i\log(iz + (1-z^2)^{1/2})$
- $\log(iz - (1-z^2)^{1/2})$
- $-i\log(iz - (1-z^2))$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

A function U is said to be harmonic if and only if _____.

Answer (Please select your correct option)

- $U_{xy} + U_{yx} = 0$
- $U_x + U_y = 0$
- $U_x^2 + U_y^2 = 0$
- $U_{xx} + U_{yy} = 0$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 6 of 33

If U and V are harmonic functions then $f(z) = U + iV$ is _____

Answer (Please select your correct option)

- analytic function.
- need not be analytic function.
- analytic function only at $z=0$.
- none of the others.

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Question Summary : (Attempted Question)
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MTH632 Complex Analysis and Differential Geometry

Question No : 4 of 33

A function $f(z)$ is analytic function if _____ part $f(z)$ is analytic.

Answer (Please select your correct option)

(a) real

(b) imaginary

(c) both (a) and (b)

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 2 of 33

The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____

Answer (Please select your correct option)

- $w = \ln r + i(\theta + 2n\pi)$
- $w = re^{i\theta}$
- $w = r + i(\theta + 2n\pi)$
- $w = \ln r + i\theta$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No: 4 (of 5)

A function $U(x, y)$ is said to be harmonic if and only if _____

Answer (Please select your correct option)

C $U_{xx} + U_{yy} = 0$

C $U_x + U_y = 0$

C $U_x^2 + U_y^2 = 0$

C $U_{xx} + U_{xy} = 0$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 21 of 33

If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$ _

Answer (Please select your correct option)

- $-\vec{b}$
- $\vec{b}$
- $\vec{t}$
- $\vec{n}$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 22 of 33

Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.

Answer (Please select your correct option)

- $(\phi + \psi)(v)$
- $(\phi + \psi)(v) = (\psi)(v)$
- $(\phi + \psi)(v) = (\phi)(v) + (\psi)(v)$
- $(\phi + \psi)(v) = (\phi)(v)$

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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 8 of 33

Any function which is analytic at a point z_0 must have a _____ series about z_0

Answer (Please select your correct option)

- Maclaurin's
- Taylor's

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Question Summary : (Attempted Question)
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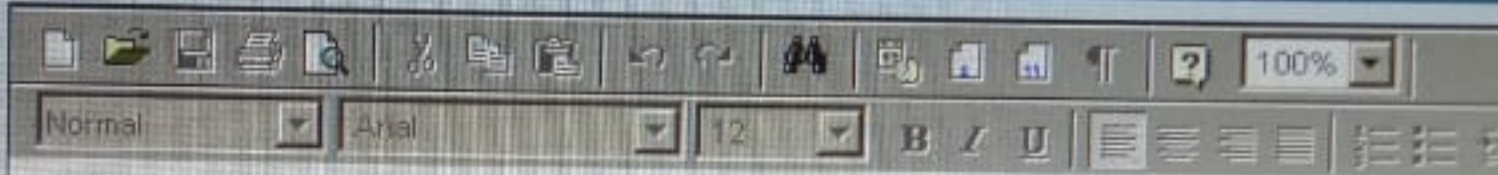
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MTH632 Complex Analysis and Differential Geometry

Question No : 30 of 33

Show that $\frac{d}{dz}(\operatorname{arcsinh}(z)) = \frac{1}{\sqrt{1+z^2}}$

Answer (Please click here to Add Answer)



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Question Summary : (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 14 of 33

The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____

Answer (Please select your correct option)

- $\int_a^b w(t)dt = \int_a^b u(t)\,v(t)dt$
- $\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
- $\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
- $\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$

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Question Summary : (Attempted Question)

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Question No : 3 of 33 Marks: 1 (Budget)

The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____

Answer (Please select your correct option)

- C i
- C -1
- C 1
- C -2

3

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MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 33

The principal value of 1^{2i} is _____

Answer (Please select your correct option)

- 0
- 1
- i
- $2i$

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Question Summary : (Attempted Question)

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Question No : 5 of 33 Marks: 1 (Budgeted Time Min)

An infinite sequence $z_1, z_2, \ldots, z_n, \ldots$ of complex numbers has a limit 'z' if, for each positive number ε, there exists a positive integer n_0 such that $|z_n - z| < \varepsilon$ whenever

_____

Answer (Please select your correct option)

- $n = n_0$
- $n < n_0$
- $n > n_0$
- $n - n_0$

Question No : 1 of 33

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____

Answer (Please select your correct option)

- harmonic in D
- equal in D
- homogeneous in D
- steady in D

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Question Summary (Unanswered Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

MTH632 Complex Analysis and Differential Geometry

Question No : 3 of 10

The logarithmic function $w = \log z$, where $z = re^{i\theta}$ and $w = u + iv$, then

Answer (Please select your correct option)

- $w = \log z + i(\theta + 2n\pi)$
- $w = re^{i\theta}$
- $w = z + i(\theta + 2n\pi)$
- $w = \log z + i\theta$

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Question Summary (Attempted Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 17 of 33

Let C be the positively oriented circle $|z| = 2$. Since the function $f(z) = \frac{z}{9 - z^2}$ is analytic within and on C and

Cauchy Integral formula $\int_C \frac{z}{(9 - z^2)(z + 1)} dz =$ _____

Answer (Please select your correct option)

- $\frac{\pi}{9}$
- $\frac{\pi}{5}$
- 2π
- $2\pi i$

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Question Summary : (Attempted Question)
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MTH632 Complex Analysis and Differential Geometry

Question No : 23 of 33

A function $F: \boldsymbol{R}^n \to \boldsymbol{R}^m$ is differentiable if all its Euclidean coordinate functions are _____

Answer (Please select your correct option)

integrable

differentiable

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Question Summary : (Attempted Question)

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Question No : 2 of 33

The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____

Answer (Please select your correct option)

- $w = \ln r + i(\theta + 2n\pi)$
- $w = re^{i\theta}$
- $w = r + i(\theta + 2n\pi)$
- $w = -\ln r + i\,\theta$

MTH632 Complex Analysis and Differential Geometry

Question No : 1 of 10

Marks: 1

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____

Answer: (Please select your correct option)

- harmonic in D
- equal in D
- homogeneous in D
- unsafe in D

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Question Summary (Attempted, Question)

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MTH632 Complex Analysis and Differential Geometry

Question No : 5 of 33

An infinite sequence $z_1, z_2, \ldots, z_n, \ldots$ of complex numbers has a limit 'z' if, for each positive number ε, there exists a p

_____

Answer (Please select your correct option)

- $n = n_0$
- $n < n_0$
- $n > n_0$
- $n - n_0$

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Question Summary : (Attempted Question)

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